Raisa Karnik

raisak@tamu.edu Linkedin

EDUCATION

Texas A&M University **August 2023 - May 2027**
Bachelor of Science, Computer Engineering
GPA: 3.7/4.0
Relevant Coursework: Machine Learning, Data Structures and Algorithms, Statistics and Probability, Discrete Mathematics, Linear Algebra, Computer Systems, Computer Architecture

RESEARCH EXPERIENCE

NSF CISE REU at UT Austin **June 2026 - August 2026**
Research Intern | Dr. Ufuk Topcu and Dr. David Fridovich-Keil

- Implemented reinforcement learning (model-free proximal policy optimization) finetuning of diffusion policies and deployed on physical simulation (Robomimic and Gymnasium, MuJoCo backend)
- Found Diffusion Policy Policy Optimization is more sample efficient than standard Gaussian Proximal Policy Optimization

Learning and Emerging Networked Systems Laboratory **August 2025 - Present**
Research Assistant | Dr. Dileep Kalathil and Dr. Srinivas Shakkottai

- Comparing different vision-language-action (VLA) architectures on manipulation tasks to understand model limitations and improve generalizability
- Finetuned and evaluated smolVLA, $\pi 0$, and GR00T on pick-and-place tasks using LeRobot and SO-101 robots
- Found that trials with multiple objects or out-of-distribution objects saw task completion drop to approximately 50% of baseline

Microsoft Research **June 2025 - August 2025**
Research Intern | Dr. Ganesh Ananthanarayanan

- Deployed smolVLA and $\pi 0$ using real image streams (PyTorch, OpenCV libraries and USB/IP) to study latency, accuracy, and size-performance tradeoffs
- Created evaluation metrics and tested accuracy and task completion, which were further used in GPU scheduling applications

Humanitarian Robotics and AI Laboratory **October 2023 - Present**
Research Assistant | Dr. Robin Murphy

- Using computer vision (PyTorch, Tensorflow, Rasterio, Shapely libraries) to annotate and model road and building damage at post-disaster events
- Optimized imagery tiling and reduced dataset pre-processing time by 4x, finetuned and tested UNet with multimodal imagery
- Conducted quantitative analysis using confusion matrices between annotated and ground truth building damage, found that labels were misaligned at increasingly severe damage
- Modeled epistemic and aleatoric uncertainty in building damage assessment, reduced expected calibration error by 90%

PUBLICATIONS

- Thomas Manzini, Priyankari Perali, **Raisa Karnik**, Mihir Godbole, Hasnat Abdullah, Robin Murphy. "Non-Uniform Spatial Alignment Errors in sUAS Imagery From Wide-Area Disasters." 2024. *International Conference on Robot & Human Interactive Communication (**RO-MAN**) 2025.*
- Thomas Manzini, Priyankari Perali, **Raisa Karnik**, Robin Murphy. "CRASAR-U-DROIDs: A Large Scale Benchmark Dataset for Building Alignment and Damage Assessment in Georectified sUAS Imagery." 2024. *arXiv preprint. arXiv:2407.17673.*
- Thomas Manzini, Priyankari Perali, **Raisa Karnik**, Robin Murphy. "A Benchmark Dataset for Spatially Aligned Road Damage Assessment in Small Uncrewed Aerial Systems Disaster Imagery." 2025. *40th Annual AAAI Conference on Artificial Intelligence **(AAAI)** 2026.*
- Thomas Manzini, Priyankari Perali, **Raisa Karnik**, Stephen Johnson, Robin R. Murphy. "Looks Can be Deceiving: Annotator and Reviewer Performance Across Imagery Sources in Crowd-Sourced Aerial Damage Assessment." 2026. *ACM Conference on Human-AI Complementarity and Alignment (**HCOMP**) 2026.*
- Thomas Manzini*, **Raisa Karnik***, Priyankari Perali, Robin R. Murphy. "Cross-Sensor Uncertainty Quantification in Drone and Satellite Post-Disaster Imagery." 2026. *Under Review at AAAI 2027.*

PRESENTATIONS

- Raisa Karnik. "CRASAR-U-DROIDs: Disaster Response Overhead Inspection Dataset." Texas Regional Robotics Symposium, April 2024. **[Poster]**
- Thomas Manzini, Priyankari Perali, Raisa Karnik, Robin Murphy. "CRASAR-U-DROIDs: A Large Scale Benchmark Dataset for Building Alignment and Damage Assessment in Georectified sUAS Imagery." MIT Lincoln Lab HADR Group, August 2024; & Military Operations Research Society, September 2024. **[Talk]**
- Raisa Karnik. "RL Finetuning of Diffusion Policies for Robot Control Tasks." UT Austin Summer Research Scholars Poster Session, July 2026. **[Poster]**

ACTIVITIES

OurCS **October 2024**

Workshop

- Hosted by Carnegie Mellon University, 3-day research focused workshop
- Presented poster on using machine learning to assess cultural differences in post-market surveillance data for CPAP masks

Society of Women in Engineering **September 2023 - May 2025**

Freshmen Leadership, Education Committee

- Service organization dedicated to empowering women in engineering academically and professionally through active community engagement
- Organized educational events to foster the development of engineering skills needed in industry and research

Jounal of Young Investigators **July 2023 - Present**

Editor

- Working with journalists and editing scientific articles about recent advancements and innovation in STEM
- Communicating with journalists, negotiating deadlines, and providing feedback on articles; works include scientific articles, press releases on research, and career feature articles

TEACHING AND MENTORING

ENGR 102: Engineering Lab I - Computation **Fall 2025**

Teaching Assistant

- Assisting during lecture and lab sessions, clarify and reinforce core concepts and help students develop problem solving skills

Matriculate **February 2024 - Present**

Advising Fellow

- Mentored high school seniors from low-income backgrounds in the college application process
- Organized applications, revised essays, and educated on financial aid packages for three high school students to help find best fit colleges
- Educating prospective applicants about the program and assisting outreach as a student liaison

SKILLS

Software

- Python, C++, MATLAB, Verilog
- PyTorch, OpenCV, NumPy, Pandas, Rasterio, Git, Linux
- Physical simulators: Robomimic, Gymnasium, Isaac Lab
- Computer vision pipelines and evaluation

Hardware

- SO-100/SO-101 manipulators
- Sensor based perception
- Breadboard prototyping and debugging
- FPGA programming